auditd Log Fields

DG Interns Hub - Week 3 | Hands-On Task 3: Explore Linux Auditing with auditd

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What I did

On my Ubuntu VM I installed auditd, enabled the service and added an audit rule that records every program run by /usr/bin/apt and tags it with the key nano:

```
sudo auditctl -a always,exit -F arch=b64 -S execve -F exe=/usr/bin/apt -k nano
```

I then ran `sudo apt install nano` and searched for it with `sudo ausearch -k nano -i` and `sudo ausearch -m EXECVE -ts recent -i`. The -i option converts numbers such as user IDs into readable names. One command produces several linked records that share the same timestamp and serial number.

Common fields in every record

Field	What it shows and why it is useful
type	The kind of record, for example SYSCALL, EXECVE, PATH, CWD, PROCTITLE or CONFIG_CHANGE. It tells me which part of the event I am looking at.
msg=audit(time: serial)	The timestamp and a serial number. Records that share the same serial belong to the same event, so I can put a full event together and place it on a timeline.

SYSCALL record (the system call that was made)

Field	What it shows and why it is useful
arch	The system architecture (x86_64). Needed to read the syscall number correctly.
syscall	The system call that triggered the event (execve means a program was started).
success / exit	Whether the call worked and its return code. A failed call (success=no) can point to a blocked or mistaken attempt.
a0 to a3	The raw arguments passed to the system call. Useful for deep technical analysis, but the readable arguments are in the EXECVE record.
items	The number of PATH records attached to this event.
ppid / pid	The parent and the process ID. Lets me trace which process started which, and follow a chain of activity.
auid	The audit user ID: the original user who logged in, which stays the same even after sudo. This is the most important field for knowing who really ran the command (here: ubuntu).
uid / gid	The user and group the process ran as (here: root, because I used sudo). Comparing it with auid shows privilege escalation.
euid, suid, fsuid, egid, sgid, fsgid	The effective, saved and file-system user and group IDs. They help detect a process that changed its privileges.
tty	The terminal the command came from (here: pts1, an SSH session). Shows whether it was run interactively or remotely.
ses	The login session number. Groups all commands from one login.
comm	The short name of the command (apt).

Field	What it shows and why it is useful
`exe`	The full path of the program that ran (/usr/bin/apt). Important for spotting a program running from an unusual location.
`subj`	The security label (SELinux or AppArmor). Shows whether the process was confined (here: unconfined).
`key`	The label I gave the rule (nano). It lets me find related events quickly with ausearch -k.

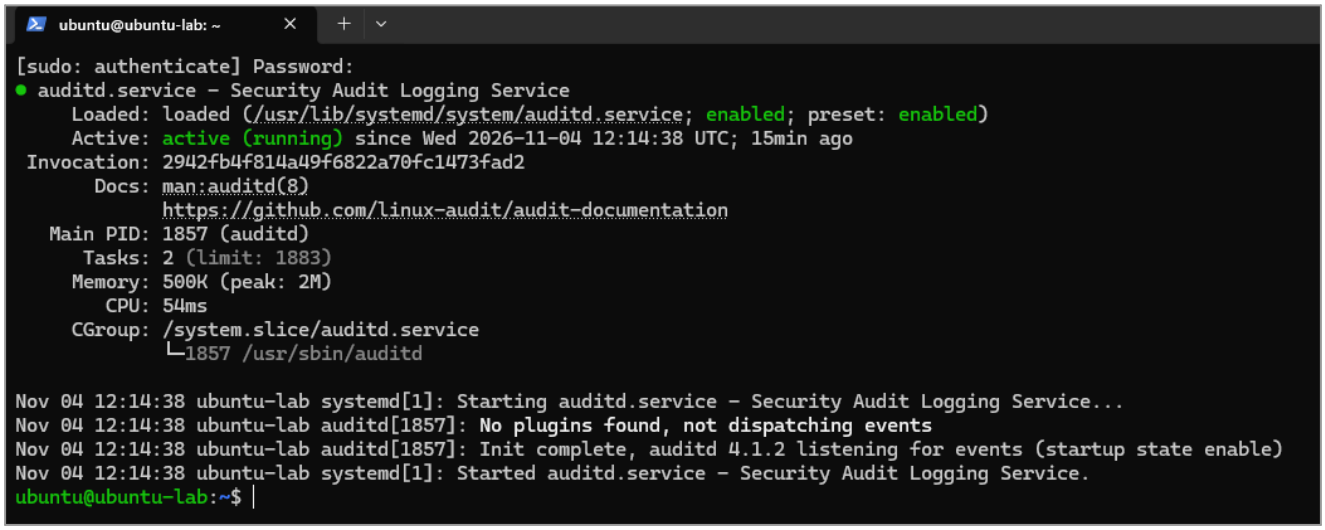
Other records in the same event

Field	What it shows and why it is useful
`EXECVE: argc, a0, a1, a2`	The actual command line, one argument per field. In my event: argc=3, a0=apt, a1=install, a2=nano, so the command was apt install nano. This is the field that answers what was run.
`PROCTITLE: proctitle`	The full command line as one string (apt install nano). A quick way to read the command.
`CWD: cwd`	The directory the command was run from (/home/ubuntu). Gives context about where the user was working.
`PATH: name, inode, mode`	The file that was run or touched, with its inode and permissions. Helps confirm the exact binary used and its file permissions.
`PATH: ouid, ogid`	The owner and group of that file (root). A file owned by an unexpected user can be a warning sign.
`CONFIG_CHANGE: op, key, res`	Recorded when the audit rules change (op=add_rule). It shows that my rule was added and succeeded (res=yes). Attackers sometimes try to change audit rules, so this record is worth monitoring.

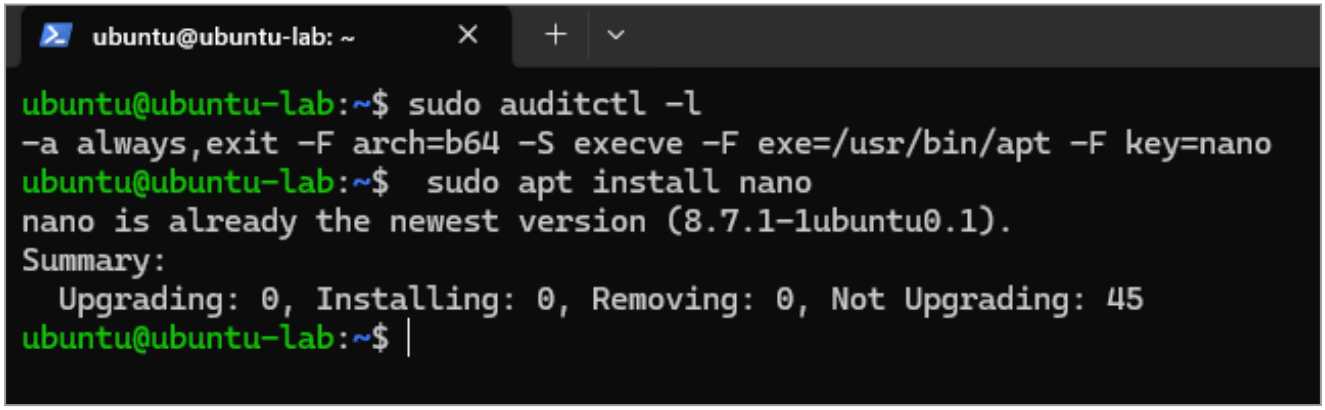
My example event

Item	Value from my log
Time	11/04/2026 12:35:26.207 (serial 701)
Command (EXECVE)	apt install nano (argc=3)
Program (exe)	/usr/bin/apt
Who (auid / uid)	auid=ubuntu, uid=root (ran with sudo)
Terminal / session	tty=pts1, ses=4
Rule key	nano

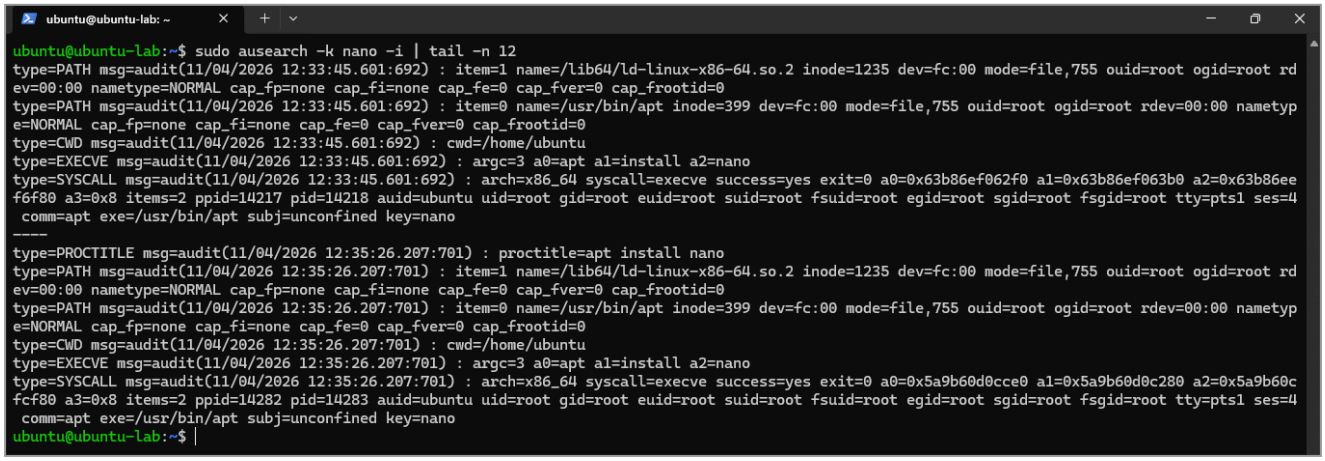
Evidence screenshots



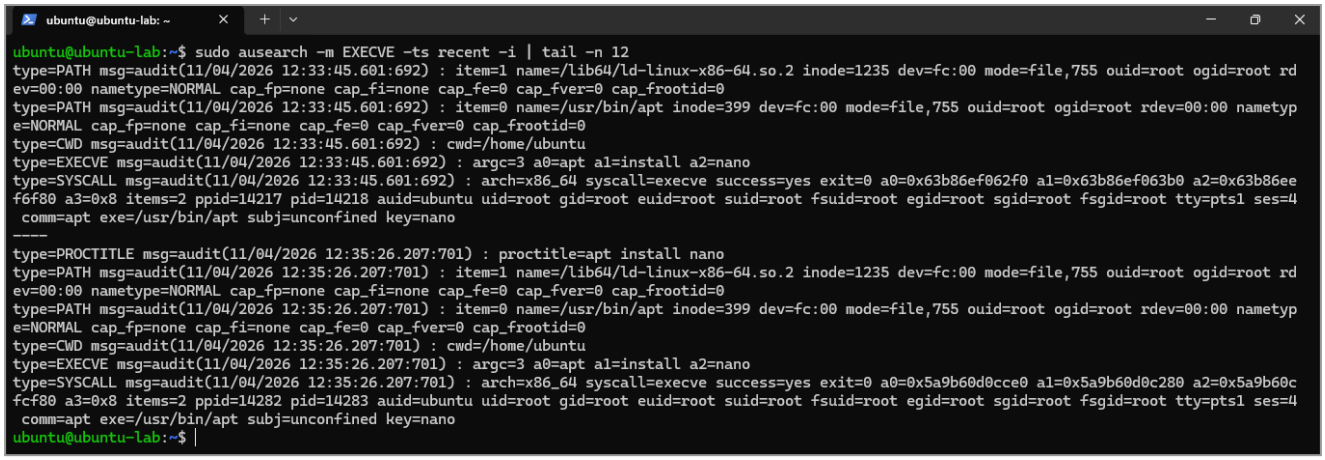
1. auditd service installed, enabled and running (sudo systemctl status auditd)



2. Audit rule loaded with key nano, then the audited command run (sudo auditctl -l, sudo apt install nano)



3. Event found by key (sudo ausearch -k nano -i)



4. Event found by message type (sudo ausearch -m EXECVE -ts recent -i)